

Barrett Larson

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EDUCATION

Virginia Tech, College of Engineering

Blacksburg, VA

Bachelor of Science in Computer Science, Minor in Mathematics | GPA: 3.70

Expected May 2027

- **Relevant Coursework:** Data Structures & Algorithms (Java), GUI Programming (JavaScript), Database Management Systems (SQL), Computer Systems (C)

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, Java, SQL, C, HTML, CSS

Frameworks/Libraries: React.js, FastAPI, Node.js, Express.js, SQLAlchemy

Databases & Cloud: PostgreSQL, MySQL, MongoDB, AWS (Cloud Practitioner)

EXPERIENCE

Expedition Technology

Herndon, VA

Software Engineering Intern

May 2026 – Present

- Architecting a real-time collaborative annotation editor over WebSockets, backed by FastAPI and PostgreSQL, with live presence, shared annotation rendering, throttled autosave, and conflict resolution via annotation-level locking.
- Implemented frontend features for an AI training data platform, including metadata-driven image boundary rendering with axis-based fallback for missing metadata.
- Validated changes across the codebase with Playwright end-to-end and Vitest unit tests, covering regression cases in existing TypeScript and React logic.
- Added server-side validation rejecting incompatible annotation types before dataset ingestion.

CARFAX

Centreville, VA

Software Developer Intern

May 2025 – August 2025

- Developed an AWS Lambda function that queried Salesforce to retrieve account and contact data, publishing results to dedicated AWS SNS topics.
- Upgraded backend codebases to Java 21 with Spring Boot 3.5+, replacing deprecated libraries and resolving runtime-breaking changes across multiple microservices to address security vulnerabilities.
- Refactored legacy Node.js Lambda functions to Python, resolving critical security vulnerabilities and improving reliability with pytest-based unit tests.

PROJECTS

HydroScript | *Python, FastAPI, React.js, TypeScript*

- Built a progressive web app that lets a household control their pool/spa systems from any phone or browser, with push notifications when the spa or pool reaches its target temperature.
- Polled the manufacturer's cloud service once in the background and served all users from a shared cache, keeping upstream load constant regardless of user count.
- Secured the app behind a login gate and added a nightly heater auto-shutoff, verified by a hardware-free test suite using simulated devices.

Fraternity Formals Ticket Platform | *MongoDB, Express.js, React.js, Node.js*

- Built a complete ticket-purchasing platform for chapter formals as a fee-free alternative to third-party ticketing services, which charge roughly \$1,000 in fees per event at chapter scale.
- Integrated Stripe for secure payment processing with a streamlined checkout flow, guest name collection, and automated email confirmations.

LEADERSHIP

Phi Delta Theta

Blacksburg, VA

Secretary

August 2024 – December 2024

- Served on the executive board, contributing to chapter-level decisions and maintaining official records of all chapter meetings.